

$X_1, \dots, X_n$ iid variables with cdf F_X , and pmf/pdf f_X
 $X_{(1)} \leq \dots \leq X_{(n)}$ corresponding order statistics.

Joint distribution of $X_{(r)}$ & $X_{(s)}$, $r < s$.

jt-cdf (for both discrete and continuous r.v.s)

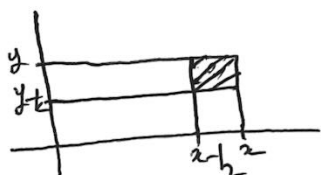
$$\begin{aligned} F_{X_{(r)}, X_{(s)}}(x, y) &= \text{Prob.}(X_{(r)} \leq x, X_{(s)} \leq y) \\ &= \text{Prob.}(\text{at least } r \text{ } X_k' \leq x, \text{ at least } s \text{ } X_k' \leq y) \\ &= \sum_{j=s}^n \sum_{i=r}^j \text{Prob.}[\text{exactly } i \text{ } X_k' \text{ in } (-\infty, x], \text{ exactly } (j-i) \text{ } X_k' \text{ in } (x, y], \\ &\quad \text{exactly } (n-j) \text{ in } (y, \infty)] \\ &= \sum_{j=s}^n \sum_{i=r}^j \frac{n!}{i!(j-i)!(n-j)!} (F_X(x))^i (F_X(y) - F_X(x))^{j-i} (1 - F_X(y))^{n-j} \text{ if } x < y \end{aligned}$$

What if $x > y$?

jt pmf of $X_{(r)}, X_{(s)}$, $r < s$ (discrete variables)

$$f_{X_{(r)}, X_{(s)}}(x, y) = F_{X_{(r)}, X_{(s)}}(x, y) - F_{X_{(r)}, X_{(s)}}(x-h, y) - F_{X_{(r)}, X_{(s)}}(x, y-k) + F_{X_{(r)}, X_{(s)}}(x-h, y-k)$$

(area in the top-right corner, h and k depend on the jump size of the discrete distribution, do not tend to 0 as in the cont. case)



jt pdf of $X_{(r)}, X_{(s)}$, $r < s$ (continuous)

$$\begin{aligned} f_{X_{(r)}, X_{(s)}}(x, y) &= \lim_{\substack{h \rightarrow 0 \\ k \rightarrow 0}} \frac{1}{hk} \text{Prob}[x < X_{(r)} \leq x+h, y < X_{(s)} \leq y+k] \\ &= \frac{n!}{(r-1)!(s-r-1)!(n-s)!} \lim_{\substack{h \rightarrow 0 \\ k \rightarrow 0}} (F_X(x))^{r-1} \left(\frac{F_X(x+h) - F_X(x)}{h} \right) (F_X(y) - F_X(x+h))^{s-r-1} \left(\frac{F_X(y+k) - F_X(y)}{k} \right) (1 - F_X(y+k))^{n-s} \\ &= C_{r,s,n} (F_X(x))^{r-1} f_X(x) (F_X(y) - F_X(x))^{s-r-1} f_X(y) (1 - F_X(y))^{n-s}, \quad -\infty < x < y < \infty. \end{aligned}$$

Jt. distribution of n ordered statistics $X_{(1)} < \dots < X_{(n)}$,

$X_1, \dots, X_n$ iid from continuous distn with cdf F_X , pdf f_X .

Jt. of $X_1, \dots, X_n$ is $f_{X_1, \dots, X_n}(x_1, \dots, x_n) = \prod_{i=1}^n f_{X_i}(x_i)$, $-\infty < x_i < \infty \forall i=1(n)$.

n transformations $X_{(1)} = g_1(X_1, \dots, X_n)$
 $\vdots$
 $X_{(n)} = g_n(X_1, \dots, X_n)$, with $n!$ inverse transformations.

Jt. pdf of $X_{(1)} < \dots < X_{(n)}$ is $f_{X_{(1)}, \dots, X_{(n)}}(x_1, \dots, x_n) = \sum_{j: \text{sum over all } n! \text{ inv. trans}} \|J_j\| \prod_{i=1}^n f_{X_i}(x_i) = n! \prod_{i=1}^n f_{X_i}(x_i)$, $-\infty < x_1 < \dots < x_n < \infty$

[each J_j (Jacobian matrix) is a permutation matrix with determinant equal to ± 1]

Moments of order statistics

$$E(X_{(r)}) = E(X_{r:n}) = \mu_{r:n} = c_{r:n} \int_{-\infty}^{\infty} x f_{r:n}(x) dx$$

(rth ordered from a sample size n)

$$f_{r:n}(x) = (F(x))^{r-1} (1-F(x))^{n-r} f(x)$$

$$c_{r:n} = \frac{1}{B(r, n-r+1)}$$

$$\text{Now, } |\mu_{r:n}| \leq \int |x| f_{r:n}(x) dx \leq \int |x| f(x) dx = E|X_1|$$

($\because$ identically distd $E|X_1|$ or any $E|X_i|$)

~~Here~~ If $E|X_1|$ is finite $E(X_1)$ ($= \mu$, say) is finite

and $\mu_{r:n}$ finite $\forall r=1(1)n; n \geq 1$ (why?).

Imp. || But if $E|X_1|$ is not finite, $E(X_1)$ is not finite, we still ~~may~~ have $\mu_{r:n}$ finite for some r . At least one $\mu_{r:n}$ has to be infinite, and that happens at the extreme(s) only. (why again?)

Ex $X \sim \text{Cauchy}(0,1)$

$$\mu_{r:n} = \frac{c_{r:n}}{\pi} \int x \left(\frac{1}{2} + \frac{1}{\pi} \tan^{-1} x\right)^{r-1} \left(\frac{1}{2} - \frac{1}{\pi} \tan^{-1} x\right)^{n-r} \frac{1}{1+x^2} dx$$

is finite for $2 \leq r \leq n-1, n \geq 2$

(easy way to check, let $u = F(x)$, then $\mu_{r:n} = \frac{c_{r:n}}{\pi} \int_0^1 x(u) u^{r-1} (1-u)^{n-r} du$,
In this case $x(u) = F^{-1}(u) = \tan \pi(u - \frac{1}{2}) = \cot \pi u$).

The r, s th order joint moment

$$E(X_{r:n} X_{s:n}) = \mu_{r,s:n} = c_{r,s:n} \int \int xy f_{r,s:n}(x,y) dx dy$$

$$= c_{r,s:n} \int \int x(u) y(v) u^{r-1} (v-u)^{s-r-1} (1-v)^{n-s} du dv$$

$$0 < u < v < 1.$$

$$\text{Cov}(X_{r:n}, X_{s:n}) = \sigma_{r,s:n} = \mu_{r,s:n} - \mu_{r:n} \mu_{s:n}$$

$$\text{Var}(X_{r:n}) = \sigma_{r,r:n} = \sigma_{r:n}^{(2)} = \mu_{r,n}^{(2)} - \mu_{r:n}^2, \text{ etc.}$$

Next, some asymptotic or large sample results (last revision topic).